Package 'levyfit'

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Title Maximum Likelihood Estimation for Levy Degradation Processes

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Description Maximum likelihood estimation of the parameters of degradation models based on Levy processes (Wiener, gamma, inverse Gaussian, variance-gamma and Meixner processes) from one or several trajectories observed at possibly irregular inspection times.

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URL https://github.com/rytracetito/levy_packages/tree/main/levyfit, https://github.com/rytracetito/levy_packages

BugReports https://github.com/rytracetito/levy_packages/issues

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`levyfit-package` *levyfit: maximum likelihood estimation for degradation processes*

Description

`levyfit` estimates the parameters of stochastic degradation models from one or several degradation trajectories observed at inspection times, which can be irregular and differ from one trajectory to another. Each estimator is the maximum likelihood estimator (MLE); the help page of each function details the main mathematical steps of its derivation.

Functions of the package

Estimation of the parameters (click on a function name for its help page):

Function	Process	Parameters	Computation
`wiener_mle()`	Wiener with drift	`mu`, `sigma2`, `s2`	closed form
`gamma_mle()`	gamma	shape `a`, rate `b`	root of one equation
`ig_mle()`	inverse Gaussian	`mu`, `eta`, `eta_unbiased`	closed form
`vg_mle()`	variance-gamma	`theta`, `sigma`, `nu`	EM, then L-BFGS-B
`meixner_mle()`	Meixner	`a`, `b`, `d`, `m`	profile likelihood

`s2` and `eta_unbiased` are the unbiased versions of `sigma2` and `eta`. Every estimation function returns the point estimates and, with `ci = TRUE`, asymptotic confidence intervals (see `wiener_mle()`). When the estimator does not exist for the data, it gives a warning and returns `NA` estimates instead of stopping.

Inspection times following a chosen pattern, and simulated trajectories to try the estimators, are given by the package `DegradSim` (`DegradSim::inspection_times()` and its simulation functions).

Data

Every estimation function accepts the data in three forms:

- one trajectory: `f(x, t)`, with `x` the degradation levels and `t` the inspection times;
- several trajectories with the same inspection times: `f(x)`, with `x` the matrix returned by the `DegradSim` simulation functions (columns `t`, `X1`, `X2`, ...);
- several trajectories, each with its own inspection times: `f(liste)`, with `liste` a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels.

The trajectories are assumed independent and with the same law. The increments are computed within each trajectory and then pooled.

Computation of the estimators

With $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, m$, the durations and increments between two consecutive inspections, pooled over the trajectories, the log-likelihood of a Lévy process is $\ell(\psi) = \sum_j \ln f_{\Delta t_j}(\Delta x_j; \psi)$.

- **Closed form**: Wiener and inverse Gaussian. The MLE is explicit, with $\hat{\mu} = S/T$, S the total degradation and T the total observed time.

- **Root of one equation**: gamma. $\hat{b} = \hat{a}T/S$ is explicit and $\hat{a}$ is the unique root of a decreasing function, found by `stats::uniroot()` from the method of moments estimator; the result does not depend on this starting point.
- **Numerical maximization**: variance-gamma and Meixner.
  - Variance-gamma: EM algorithm, the gamma time change being the hidden variable (each step is explicit or a unique root), started from the Wiener MLE, then L-BFGS-B (`stats::optim()`) with a numerical gradient.
  - Meixner: L-BFGS-B on the likelihood profiled in b ($\hat{b}$ is explicit for given a, d, m), with an explicit gradient and several starting points (`n_start`) built from the empirical mean and variance; the best interior maximum is kept.

When the supremum of the likelihood is reached at the boundary of the parameter space (for instance the Brownian limit of the variance-gamma process), the MLE does not exist: the function gives a warning and returns `NA`. The help page of each function details its derivation.

Confidence intervals

With `ci = TRUE`, the intervals are Wald intervals. The observed Fisher information $J = -\nabla^2 \ell(\hat{\psi})$ is computed numerically by `stats::optimHess()` for every model, and the standard error of $\hat{\psi}_k$ is $\widehat{se}_k = \sqrt{(J^{-1})_{kk}}$. With z the quantile of order $(1 + level)/2$ of the standard normal distribution:

- real parameter (`mu` of the Wiener process, `theta` of the variance-gamma process, `b` and `m` of the Meixner process): $\hat{\psi}_k \pm z\, \widehat{se}_k$;
- positive parameter (all the others): the interval $\ln \hat{\psi}_k \pm z\, \widehat{se}_k / \hat{\psi}_k$ for $\ln \psi_k$ (delta method), transformed back: $[\hat{\psi}_k e^{-z\widehat{se}_k/\hat{\psi}_k},\ \hat{\psi}_k e^{z\widehat{se}_k/\hat{\psi}_k}]$, which always stays positive.

For the Wiener, inverse Gaussian and gamma processes, the numerical information agrees with its closed form; for instance, for the Wiener process, $\widehat{se}(\hat{\mu}) = \hat{\sigma}/\sqrt{T}$ and $\widehat{se}(\hat{\sigma}^2) = \hat{\sigma}^2\sqrt{2/m}$.

Parametrizations

- Wiener: $X_t - X_s \sim \mathcal{N}(\mu(t-s), \sigma^2(t-s))$.
- Gamma: $X_t - X_s \sim \Gamma(a(t-s), b)$, shape $a(t-s)$ and **rate** b, so that $E[X_t] = at/b$. The scale is $1/b$.
- Inverse Gaussian: $X_t - X_s \sim IG(\mu(t-s), \eta(t-s)^2)$.
- Variance-gamma: $X_s = \theta\gamma_s + \sigma W_{\gamma_s}$, with γ_s of mean s and variance νs.
- Meixner: $X_s \sim Meixner(a, b, ds, ms)$.

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References

Gamma process: van Noortwijk, J. M. (2009). A survey of the application of gamma processes in maintenance. *Reliability Engineering & System Safety*, 94(1), 2-21. doi:10.1016/j.ress.2007.03.019

Wiener process: Whitmore, G. A. (1995). Estimating degradation by a Wiener diffusion process subject to measurement error. *Lifetime Data Analysis*, 1, 307-319.

Inverse Gaussian process: Wang, X. and Xu, D. (2010). An inverse Gaussian process model for degradation data. *Technometrics*, 52(2), 188-197. doi:10.1198/TECH.2009.08197

Variance-gamma process: Madan, D. B., Carr, P. P. and Chang, E. C. (1998). The variance gamma process and option pricing. *European Finance Review*, 2(1), 79-105. doi:10.1023/A:1009703431535

Meixner process: Schoutens, W. and Teugels, J. L. (1998). Lévy processes, polynomials and martingales. *Communications in Statistics. Stochastic Models*, 14(1-2), 335-349.

The references of each method are given in the help page of each function.

See Also

Useful links:

- https://github.com/rytracetito/levy_packages/tree/main/levyfit
- https://github.com/rytracetito/levy_packages
- Report bugs at https://github.com/rytracetito/levy_packages/issues

Examples

```
# 1. Inspection times: 10 trajectories inspected more and more often
plan <- DegradSim::inspection_times("increasing", n = 20, n_traj = 10, tmax = 100)

# 2. Simulated data, here a gamma process (a = 2, b = 3)
set.seed(1)
x <- DegradSim::sim_gamma(plan, a = 2, b = 3)

# 3. Estimation of several models on the same data
gamma_mle(x)
wiener_mle(x)
ig_mle(x)

# 4. Estimates with 95 % confidence intervals
gamma_mle(x, ci = TRUE)
```

`gamma_mle` *Maximum likelihood estimation for the gamma process*

Description

Computes the maximum likelihood estimator (MLE) of the shape a and the rate b of the homogeneous gamma process from one or several trajectories observed at possibly irregular inspection times. The MLE always exists (with probability one), is unique, and is the root of an equation in one variable.

Usage

```
gamma_mle(x, t = NULL, ci = FALSE, level = 0.95)
```

Arguments

x	Degradation levels of one trajectory, with strictly positive increments; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly with `increments_data()`.
t	Inspection times of the trajectory, strictly increasing. Not used when x is a list.
ci	FALSE (default) for the point estimates only; TRUE to add asymptotic confidence intervals (see the section below).
level	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when ci = TRUE.

Value

A named vector with a $= \hat{a}$ and b $= \hat{b}$.

With ci = TRUE, a data frame with one row per parameter and the columns parameter, estimate, lower and upper (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are NA, with a warning.

Model and data

The gamma process is increasing, with independent increments $X_t - X_s \sim \Gamma(a(t-s), b)$ (shape $a(t-s)$, rate b), so that $E[X_t] = at/b$ and $Var(X_t) = at/b^2$. The trajectories are independent and have the same law; each one is observed at its own inspection times. Let $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, m$, be the durations and degradation increments ($\Delta x_j > 0$) between two consecutive inspections of a same trajectory, gathered over all the trajectories, $T = \sum_j \Delta t_j$, $S = \sum_j \Delta x_j$, $U = \sum_j \Delta t_j \ln \Delta x_j$, the speeds $r_j = \Delta x_j / \Delta t_j$ and the weights $w_j = \Delta t_j / T$.

Likelihood

The increments being independent, the log-likelihood is

$$\ell(a, b) = \sum_j \left[(a\Delta t_j - 1) \ln \Delta x_j - \ln \Gamma(a\Delta t_j) + a\Delta t_j \ln b - b\Delta x_j \right].$$

It is a two-parameter exponential family with sufficient statistic (U, S).

Maximum likelihood estimator

1. The score in b is $\partial\ell/\partial b = aT/b - S$: for a fixed a, ℓ is maximal at $\hat{b}(a) = aT/S$.
2. The profile log-likelihood $\ell_p(a) = \ell(a, \hat{b}(a))$ has derivative

$$g(a) = U - \sum_j \Delta t_j \, \psi(a\Delta t_j) + T \ln \frac{aT}{S},$$

   where ψ is the digamma function.
3. Since $\psi'(x) > 1/x$ for $x > 0$, g is strictly decreasing. Moreover $g(a) \to +\infty$ as $a \to 0$ and, by Jensen's inequality, $g(a) \to T\big(\sum_j w_j \ln r_j - \ln \sum_j w_j r_j\big) < 0$ as $a \to \infty$, unless all the speeds are equal.
4. Hence g has a unique root $\hat{a}$, and $\hat{b} = \hat{a}T/S$. If all the speeds are equal (in particular if $m = 1$), the MLE does not exist (NA with a warning); this has probability zero when $m \geq 2$.

The root is computed by `stats::uniroot()` in $\ln a$, starting from the method of moments: with $\hat{r} = S/T$ and $\hat{\sigma}^2 = \sum_j (\Delta x_j - \hat{r}\Delta t_j)^2 / (T - \sum_j \Delta t_j^2 / T)$, $a_{MM} = \hat{r}^2/\hat{\sigma}^2$. Since g is monotone, the search converges whatever the starting value.

Bias

No unbiased version of $\hat{a}$ and $\hat{b}$ is available. With few increments (about ten) both are biased upwards, by about 40 percent; the bias vanishes as the number of increments grows. Their ratio $\hat{a}/\hat{b} = S/T$ is unbiased.

Confidence intervals and estimation failures

With `ci = TRUE`, the function also returns asymptotic confidence intervals. Let $\hat{\psi}$ be the MLE and ℓ the log-likelihood. The observed Fisher information $J = -\partial^2\ell/\partial\psi\partial\psi^\top$ is computed at $\hat{\psi}$ by numerical differentiation (`stats::optimHess()`), and the standard errors are the square roots of the diagonal of J^{-1}. With z the quantile of order $(1 + level)/2$ of the standard normal distribution, the interval of a parameter ψ_k is

- $\hat{\psi}_k \pm z\, se_k$ for a real parameter (`mu` of the Wiener process, `theta` of the variance-gamma process, `b` and `m` of the Meixner process);
- $\hat{\psi}_k \exp(\pm z\, se_k/\hat{\psi}_k)$ for a positive parameter: the interval $\ln\hat{\psi}_k \pm z\, se_k/\hat{\psi}_k$ for $\ln\psi_k$, transformed back, which always stays positive.

These intervals are asymptotic: their coverage is close to `level` when the number of increments is large. The unbiased estimates `s2` (Wiener) and `eta_unbiased` (inverse Gaussian) are given without interval. When J is not positive definite (estimate close to a boundary of the parameter space), the bounds are `NA`.

When the maximum likelihood estimator does not exist for the data (for instance all the speeds $\Delta x_j/\Delta t_j$ equal, or a maximum at a boundary of the parameter space), the function does not stop: it gives a warning with the reason and returns `NA` estimates, with the reason in the attribute `"message"`. Invalid data (times not increasing, decreasing levels for an increasing process, too few observations) still give an error.

References

van Noortwijk, J. M. (2009). A survey of the application of gamma processes in maintenance. *Reliability Engineering & System Safety*, 94(1), 2-21. doi:10.1016/j.ress.2007.03.019

See Also

`wiener_mle()`, `ig_mle()`

Examples

```
# 1. One trajectory inspected every time unit up to 50 (a = 2, b = 3)
t <- seq(0, 50, by = 1)
set.seed(1)
x <- DegradSim::sim_gamma(t, a = 2, b = 3)
gamma_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
gamma_mle(x[, "X1"], t)

# 2. One trajectory with irregular inspections, more and more frequent
t <- DegradSim::inspection_times("increasing", n = 50, tmax = 50)$time
x <- DegradSim::sim_gamma(t, a = 2, b = 3)
gamma_mle(x)
```

```
# 3. Ten trajectories with the same inspection times
t <- seq(0, 50, by = 5)
x <- DegradSim::sim_gamma(t, a = 2, b = 3, n_traj = 10)
gamma_mle(x)

# 4. Ten trajectories, each with its own inspection times
plan <- DegradSim::inspection_times("decreasing", n = 5:14, n_traj = 10, tmax = 50)
x <- DegradSim::sim_gamma(plan, a = 2, b = 3)
gamma_mle(x)

# 5. Estimates with 95 % confidence intervals
gamma_mle(x, ci = TRUE)

# 6. Your own data: a list of data frames, one per trajectory,
#    with the times in the first column and the levels in the second
liste <- list(data.frame(t = c(0, 10, 20, 30), X = c(0, 6.1, 13.4, 19.8)),
              data.frame(t = c(0, 15, 30), X = c(0, 9.7, 20.5)))
gamma_mle(liste)

# The gamma process only increases: a decrease stops the estimation
try(gamma_mle(c(0, 2, 1.5, 3), c(0, 1, 2, 3)))
```

`ig_mle` *Maximum likelihood estimation for the inverse Gaussian process*

Description

Computes the maximum likelihood estimator (MLE) of the parameters μ and η of the homogeneous inverse Gaussian (IG) process from one or several trajectories observed at possibly irregular inspection times. The MLE is obtained in closed form by maximizing the likelihood directly, without numerical optimization.

Usage

```
ig_mle(x, t = NULL, ci = FALSE, level = 0.95)
```

Arguments

`x`	Degradation levels of one trajectory, with strictly positive increments; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly with `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals (see the section below).
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.

Value

A named vector with the maximum likelihood estimates `mu` $= \hat{\mu}$ and `eta` $= \hat{\eta}$, and the unbiased estimate `eta_unbiased` $= \tilde{\eta}$ of η.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Model and data

$IG(m, \lambda)$ denotes the inverse Gaussian distribution with mean m and shape λ, with density

$$\sqrt{\frac{\lambda}{2\pi x^3}}\exp\Big(-\frac{\lambda(x-m)^2}{2m^2x}\Big), \qquad x>0.$$

The IG process is increasing, with independent increments $X_t - X_s \sim IG(\mu(t-s), \eta(t-s)^2)$, so that $E[X_t] = \mu t$ and $Var(X_t) = \mu^3 t/\eta$. The trajectories are independent and have the same law; each one is observed at its own inspection times. Let $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, m$, be the durations and degradation increments ($\Delta x_j > 0$) between two consecutive inspections of a same trajectory, gathered over all the trajectories, $T = \sum_j \Delta t_j$ and $S = \sum_j \Delta x_j$.

Likelihood

With $\delta = 1/\mu$, the log-likelihood is

$$\ell(\mu,\eta) = c_0 + \frac{m}{2}\ln\eta - \frac{\eta}{2}\sum_j \frac{(\delta\Delta x_j - \Delta t_j)^2}{\Delta x_j}, \qquad c_0 = \sum_j \Big[\ln\Delta t_j - \frac{1}{2}\ln(2\pi) - \frac{3}{2}\ln\Delta x_j\Big].$$

Maximum likelihood estimator

1. Expanding the square gives
$$\sum_j \frac{(\delta\Delta x_j - \Delta t_j)^2}{\Delta x_j} = \frac{(\delta S - T)^2}{S} + \hat{Q}, \qquad \hat{Q} = \sum_j \frac{\Delta t_j^2}{\Delta x_j} - \frac{T^2}{S} \geq 0,$$
with $\hat{Q} = 0$ if and only if the speeds $\Delta x_j/\Delta t_j$ are all equal.
2. For every η, ℓ is therefore maximal at $\delta = T/S$, that is $\hat{\mu} = S/T$.
3. The remaining function $\frac{m}{2}\ln\eta - \frac{\eta}{2}\hat{Q}$ is maximal at $\hat{\eta} = m/\hat{Q}$.
4. The MLE exists and is unique if and only if $\hat{Q} > 0$, which has probability one as soon as $m \geq 2$. Otherwise the likelihood is unbounded (`NA` with a warning).

Bias

For $m \geq 2$,

$$\hat{\mu} \sim IG(\mu, \eta T), \qquad \eta\hat{Q} \sim \chi^2_{m-1},$$

and $\hat{Q}$ is independent of S: the conditional density of the increments given S does not depend on μ, and its exponent only involves $\hat{Q}$ (this extends a result of Tweedie (1957) to irregular inspection times). Hence $\hat{\mu}$ is unbiased, while, since $E[1/V] = 1/(k-2)$ for $V \sim \chi^2_k$, $E[\hat{\eta}] = \frac{m}{m-3}\eta$ for $m > 3$: $\hat{\eta}$ is biased upwards. The function also returns its unbiased version

$$\tilde{\eta} = \frac{m-3}{\hat{Q}} = \frac{m-3}{m}\hat{\eta},$$

which requires at least four increments (`NA` otherwise).

Confidence intervals and estimation failures

With `ci = TRUE`, the function also returns asymptotic confidence intervals. Let $\hat{\psi}$ be the MLE and ℓ the log-likelihood. The observed Fisher information $J = -\partial^2\ell/\partial\psi\partial\psi^\top$ is computed at $\hat{\psi}$ by numerical differentiation (`stats::optimHess()`), and the standard errors are the square roots of the diagonal of J^{-1}. With z the quantile of order $(1+level)/2$ of the standard normal distribution, the interval of a parameter ψ_k is

- $\hat{\psi}_k \pm z\, se_k$ for a real parameter (`mu` of the Wiener process, `theta` of the variance-gamma process, `b` and `m` of the Meixner process);
- $\hat{\psi}_k \exp(\pm z\, se_k/\hat{\psi}_k)$ for a positive parameter: the interval $\ln\hat{\psi}_k \pm z\, se_k/\hat{\psi}_k$ for $\ln\psi_k$, transformed back, which always stays positive.

These intervals are asymptotic: their coverage is close to `level` when the number of increments is large. The unbiased estimates `s2` (Wiener) and `eta_unbiased` (inverse Gaussian) are given without interval. When J is not positive definite (estimate close to a boundary of the parameter space), the bounds are `NA`.

When the maximum likelihood estimator does not exist for the data (for instance all the speeds $\Delta x_j/\Delta t_j$ equal, or a maximum at a boundary of the parameter space), the function does not stop: it gives a warning with the reason and returns `NA` estimates, with the reason in the attribute `"message"`. Invalid data (times not increasing, decreasing levels for an increasing process, too few observations) still give an error.

References

Tweedie, M. C. K. (1957). Statistical properties of inverse Gaussian distributions. I. *The Annals of Mathematical Statistics*, 28(2), 362-377. doi:10.1214/aoms/1177706964

Wang, X. and Xu, D. (2010). An inverse Gaussian process model for degradation data. *Technometrics*, 52(2), 188-197. doi:10.1198/TECH.2009.08197

Ye, Z.-S. and Chen, N. (2014). The inverse Gaussian process as a degradation model. *Technometrics*, 56(3), 302-311. doi:10.1080/00401706.2013.830074

See Also

`wiener_mle()`, `gamma_mle()`

Examples

```
# 1. One trajectory inspected every time unit up to 50 (mu = 0.5, eta = 2)
t <- seq(0, 50, by = 1)
set.seed(1)
x <- DegradSim::sim_ig(t, mu = 0.5, eta = 2)
ig_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
ig_mle(x[, "X1"], t)

# 2. One trajectory with irregular inspections, more and more frequent
t <- DegradSim::inspection_times("increasing", n = 50, tmax = 50)$time
x <- DegradSim::sim_ig(t, mu = 0.5, eta = 2)
ig_mle(x)

# 3. Ten trajectories with the same inspection times
```

```
t <- seq(0, 50, by = 5)
x <- DegradSim::sim_ig(t, mu = 0.5, eta = 2, n_traj = 10)
ig_mle(x)

# 4. Ten trajectories, each with its own inspection times
plan <- DegradSim::inspection_times("decreasing", n = 5:14, n_traj = 10, tmax = 50)
x <- DegradSim::sim_ig(plan, mu = 0.5, eta = 2)
ig_mle(x)

# 5. Estimates with 95 % confidence intervals
ig_mle(x, ci = TRUE)

# 6. Your own data: a list of data frames, one per trajectory,
#    with the times in the first column and the levels in the second
liste <- list(data.frame(t = c(0, 10, 20, 30), X = c(0, 5.2, 9.8, 15.3)),
              data.frame(t = c(0, 15, 30), X = c(0, 7.4, 14.6)))
ig_mle(liste)

# The inverse Gaussian process only increases: a decrease stops the estimation
try(ig_mle(c(0, 2, 1.5, 3), c(0, 1, 2, 3)))
```

`increments_data` *Data given by their increments*

Description

Builds the data of the estimation functions (`gamma_mle()`, `wiener_mle()`, `ig_mle()`, `vg_mle()`, `meixner_mle()`) from increments instead of degradation levels: the increments Δx_j over the durations Δt_j of one or several trajectories. The likelihood of these processes depends on the data only through the increments.

Usage

```
increments_data(dx, dt = 1)
```

Arguments

`dx`	Numeric vector of increments.
`dt`	Durations of the increments: one positive number (all the increments over the same duration, 1 by default) or a vector of the same length as `dx`.

Details

Giving the increments directly keeps all their precision. Levels have about 16 significant digits, so an increment smaller than about 10^{-16} times the level is rounded to 0 when computed as a difference of levels; this happens for instance with the increments reconstructed by the bridges of the package `levyBridge` on a fine grid.

Value

An object of class `"levy_increments"`, to be given as argument `x` of an estimation function (without `t`).

Examples

```
set.seed(1)
# Gamma increments over a time step 0.1, some of them very small
dx <- rgamma(200, shape = 2 * 0.1, rate = 3)
min(dx)
gamma_mle(increments_data(dx, dt = 0.1))

# Same estimate as from the levels, when no increment is lost
gamma_mle(c(0, cumsum(dx)), seq(0, 20, by = 0.1))

# Increments over different durations
dt <- runif(50, 0.5, 2)
wiener_mle(increments_data(rnorm(50, 0.5 * dt, 1.5 * sqrt(dt)), dt))
```

`meixner_mle` *Maximum likelihood estimation for the Meixner process*

Description

Computes the maximum likelihood estimator of the four parameters (a, b, d, m) of the Meixner process from one or several trajectories observed at possibly irregular inspection times. The likelihood is unbounded, so the estimator is defined as the interior local maximum of largest likelihood.

Usage

```
meixner_mle(x, t = NULL, ci = FALSE, level = 0.95, n_start = 5)
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly with `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals (see the section below).
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.
`n_start`	Number of starting points.

Value

A named vector with the estimates `a`, `b`, `d` and `m`.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Model and data

For $a > 0$, $-\pi < b < \pi$, $d > 0$ and real m, the $Meixner(a, b, d, m)$ distribution has density

$$f(x) = \frac{(2\cos(b/2))^{2d}}{2a\pi\,\Gamma(2d)} \exp\Big(\frac{b(x-m)}{a}\Big)\Big|\Gamma\Big(d + i\frac{x-m}{a}\Big)\Big|^2, \qquad x \in R.$$

The Meixner process is the Lévy process such that $X_s \sim Meixner(a, b, ds, ms)$. Its mean and variance per unit of time are $\kappa_1 = m + ad\tan(b/2)$ and $\kappa_2 = a^2 d/(2\cos^2(b/2))$. The trajectories are independent and have the same law; each one is observed at its own inspection times. Let $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, N$, be the durations and degradation increments between two consecutive inspections of a same trajectory, gathered over all the trajectories, $T = \sum_j \Delta t_j$, $S = \sum_j \Delta x_j$, $s_j = d\Delta t_j$ and $y_j = (\Delta x_j - m\Delta t_j)/a$.

Likelihood and score

The increments being independent,

$$\ell(a, b, d, m) = \sum_j \big[2s_j \ln(2\cos(b/2)) - \ln(2a\pi) - \ln\Gamma(2s_j) + by_j + \ln|\Gamma(s_j + iy_j)|^2\big].$$

With $\psi_j = \psi(s_j + iy_j)$ (digamma function of a complex argument), the score is explicit: $\partial_a \ell = \sum_j(-1 - by_j + 2y_j\Im\psi_j)/a$, $\partial_b \ell = \sum_j(y_j - s_j\tan(b/2))$, $\partial_d \ell = \sum_j \Delta t_j(2\ln(2\cos(b/2)) - 2\Re\psi(2s_j) + 2\Re\psi_j)$ and $\partial_m \ell = \sum_j \Delta t_j(2\Im\psi_j - b)/a$.

Profile in b

For fixed (a, d, m), ℓ is strictly concave in b, with unique maximizer

$$\hat{b}(a, d, m) = 2\arctan\frac{S - mT}{adT},$$

which matches the expected total increment $\kappa_1 T$ with S. The maximization is therefore done in dimension 3, on the profile log-likelihood $\ell_p(a, d, m) = \ell(a, \hat{b}(a, d, m), d, m)$.

The likelihood is unbounded

As $a \to 0$ with $b = \pi - \lambda a$, the Meixner distribution tends to a shifted gamma distribution. Placing the shift m on the smallest observed speed $\Delta x_j/\Delta t_j$ and taking $2d\Delta t_j < 1$ makes $\ell \to +\infty$: the global MLE never exists. The estimator is the interior local maximum (root of the score with negative definite Hessian) of largest likelihood. Solutions with $\pi - |b| < 0.1$ (shifted gamma limit), with $d\max_j \Delta t_j > 1000$ (Gaussian limit) or on a bound are discarded as degenerate.

Algorithm

1. Compute the mean $c_1 = S/T$ and the variance $c_2 = \sum_j(\Delta x_j - c_1\Delta t_j)^2/(T - \sum_j \Delta t_j^2/T)$ per unit of time.
2. Draw `n_start` starting points: b_0 in $(-1.5, 1.5)$ and d_0 around $1/\overline{\Delta t}$ (the first one is $b_0 = 0$, $d_0 = 1/\overline{\Delta t}$), then $a_0 = \sqrt{2c_2\cos^2(b_0/2)/d_0}$ and $m_0 = c_1 - a_0 d_0\tan(b_0/2)$, so that the starting point matches c_1 and c_2.
3. From each start, maximize ℓ_p by L-BFGS-B on $(\ln a, \ln d, m)$ with the exact gradient and wide bounds relative to the scale of the data.
4. Discard the degenerate solutions and keep the one of largest likelihood; a warning is given if a degenerate solution had a larger likelihood.

The parameters a and d are strongly correlated and d is poorly identified: hundreds of increments are needed for a stable estimation, and with few increments there is often no interior maximum, in which case the function returns `NA` with a warning.

Confidence intervals and estimation failures

With `ci = TRUE`, the function also returns asymptotic confidence intervals. Let $\hat{\psi}$ be the MLE and ℓ the log-likelihood. The observed Fisher information $J = -\partial^2\ell/\partial\psi\partial\psi^\top$ is computed at $\hat{\psi}$ by numerical differentiation (`stats::optimHess()`), and the standard errors are the square roots of the diagonal of J^{-1}. With z the quantile of order $(1+level)/2$ of the standard normal distribution, the interval of a parameter ψ_k is

- $\hat{\psi}_k \pm z\, se_k$ for a real parameter (`mu` of the Wiener process, `theta` of the variance-gamma process, `b` and `m` of the Meixner process);
- $\hat{\psi}_k \exp(\pm z\, se_k/\hat{\psi}_k)$ for a positive parameter: the interval $\ln\hat{\psi}_k \pm z\, se_k/\hat{\psi}_k$ for $\ln\psi_k$, transformed back, which always stays positive.

These intervals are asymptotic: their coverage is close to `level` when the number of increments is large. The unbiased estimates `s2` (Wiener) and `eta_unbiased` (inverse Gaussian) are given without interval. When J is not positive definite (estimate close to a boundary of the parameter space), the bounds are `NA`.

When the maximum likelihood estimator does not exist for the data (for instance all the speeds $\Delta x_j/\Delta t_j$ equal, or a maximum at a boundary of the parameter space), the function does not stop: it gives a warning with the reason and returns `NA` estimates, with the reason in the attribute `"message"`. Invalid data (times not increasing, decreasing levels for an increasing process, too few observations) still give an error.

References

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See Also

`vg_mle()`

Examples

```
# The Meixner model needs many small increments: over long intervals its
# increments are almost Gaussian and its parameters are hard to separate.

# 1. One trajectory inspected every time unit up to 500
#    (a = 1, b = 0.3, d = 1.5, m = 0.2)
t <- seq(0, 500, by = 1)
set.seed(1)
x <- DegradSim::sim_meixner(t, a = 1, b = 0.3, d = 1.5, m = 0.2)
meixner_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
meixner_mle(x[, "X1"], t)

# 2. Twenty trajectories with the same inspection times
t <- seq(0, 20, by = 1)
x <- DegradSim::sim_meixner(t, a = 1, b = 0.3, d = 1.5, m = 0.2, n_traj = 20)
meixner_mle(x)
```

```
# 3. Estimates with 95 % confidence intervals
meixner_mle(x, ci = TRUE)

# 4. Few increments over long intervals: the maximum is often at the
#    boundary (Gaussian or shifted gamma limit): warning and NA estimates
y <- DegradSim::sim_meixner(seq(0, 100, by = 10), a = 1, b = 0.3, d = 1.5, m = 0.2)
meixner_mle(y)
```

`vg_mle` *Maximum likelihood estimation for the variance-gamma process*

Description

Computes the maximum likelihood estimator (MLE) of the parameters (θ, σ, ν) of the variance-gamma (VG) process from one or several trajectories observed at possibly irregular inspection times. There is no closed form: the MLE is computed by an EM algorithm, whose steps are explicit or reduce to an equation with a unique root, followed by a quasi-Newton refinement.

Usage

```
vg_mle(x, t = NULL, ci = FALSE, level = 0.95)
```

Arguments

`x`	Degradation levels of one trajectory, with non-zero increments; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly with `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals (see the section below).
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.

Value

A named vector with the maximum likelihood estimates `theta`, `sigma` and `nu`.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Model and data

Let γ be a gamma process such that γ_s has mean s and variance νs (shape s/ν, scale ν), and W an independent standard Brownian motion. The VG process is $X_s = \theta\gamma_s + \sigma W_{\gamma_s}$, a Lévy process that can decrease. With $\lambda_s = s/\nu$, $c = \sqrt{\theta^2 + 2\sigma^2/\nu}$ and $\kappa = c/\sigma^2$, X_s has the density

$$f_s(x) = \frac{2e^{\theta x/\sigma^2}}{\sqrt{2\pi}\,\sigma\,\nu^{\lambda_s}\Gamma(\lambda_s)}\left(\frac{|x|}{c}\right)^{\lambda_s - 1/2} K_{|\lambda_s - 1/2|}(\kappa|x|), \qquad x \neq 0,$$

where K_α is the modified Bessel function of the second kind. The trajectories are independent and have the same law; each one is observed at its own inspection times. Let $(\Delta t_j, \Delta x_j)$, $j = 1, \dots, m$, be the durations and degradation increments ($\Delta x_j \neq 0$) between two consecutive inspections of a same trajectory, gathered over all the trajectories.

Likelihood

The increments being independent, $\ell(\theta, \sigma, \nu) = \sum_j \ln f_{\Delta t_j}(\Delta x_j)$. The derivatives of K_α with respect to its order have no closed form, so the score equations cannot be solved explicitly.

Boundary behaviour

As $\nu \to 0$, $\ell(\theta, \sigma, \nu)$ tends to the log-likelihood of the Wiener process with drift θ and diffusion σ; as $\nu \to \infty$, $\ell \to -\infty$. The supremum of the likelihood can therefore be reached at the Brownian limit $\nu = 0$: the MLE does not always exist (about one case in four with a single trajectory of ten increments). The function then returns `NA` with a warning and the Wiener model, see `wiener_mle()`, should be used.

As $\sigma \to 0$, $X_s \to \theta\gamma_s$: if $\theta > 0$, the VG process tends to the gamma process with shape $a = 1/\nu$ and rate $b = 1/(\theta\nu)$, and when all the increments are positive ℓ tends to the gamma log-likelihood. Hence $\sup \ell \geq \ell_\Gamma(\hat{a}, \hat{b})$, the maximum of the gamma log-likelihood. When all the increments are positive and the VG estimate does not exceed this value, the supremum is reached at the gamma limit $\sigma = 0$: the function returns `NA` with a warning and the gamma model, see `gamma_mle()`, should be used. This typically happens with monotone degradation data.

EM algorithm

Let G_j be the increment of the hidden gamma time on the j-th interval. Given Δx_j, G_j follows a generalized inverse Gaussian distribution $GIG(p_j, \chi_j, \psi)$ with $p_j = \Delta t_j/\nu - 1/2$, $\chi_j = \Delta x_j^2/\sigma^2$ and $\psi = \theta^2/\sigma^2 + 2/\nu$, whose moments $e_j^+ = E[G_j]$, $e_j^- = E[1/G_j]$ and $l_j = E[\ln G_j]$ are ratios of Bessel functions.

1. E step: compute e_j^+, e_j^-, l_j at the current parameters.
2. M step for (θ, σ^2), explicit and unique: with $S = \sum_j \Delta x_j$ and $E^+ = \sum_j e_j^+$,

$$\theta = \frac{S}{E^+}, \qquad \sigma^2 = \frac{1}{m}\Big(\sum_j \Delta x_j^2 e_j^- - \frac{S^2}{E^+}\Big).$$

3. M step for ν: with $a = 1/\nu$, the function

$$q(a) = \sum_j \left[\Delta t_j l_j - e_j^+ + \Delta t_j \ln a + \Delta t_j - \Delta t_j \psi(a\Delta t_j)\right]$$

   (ψ the digamma function) is strictly decreasing, from $+\infty$ to a negative limit: its unique root gives ν.

4. Repeat (at least 10 iterations) until the EM slows down, that is until an iteration increases ℓ by less than $10^{-6}(|\ell| + 1)$.

Each iteration increases the likelihood (Dempster, Laird and Rubin, 1977), but the EM converges slowly when ν is small (near the Brownian limit), where it could need thousands of iterations. The maximization is therefore finished by L-BFGS-B on $(\theta, \ln\sigma, \ln\nu)$, started at the last EM iterate, with a tight convergence tolerance: the likelihood and the MLE are the same, much faster. If this refinement fails or does not increase the likelihood (for instance with data on a very small scale), the

EM iterate is returned. Starting values: the Wiener MLE for (θ, σ) and $\nu = T/m$. The algorithm stops at the Brownian limit when $\nu < \max_j \Delta t_j/300$, during the EM or at the end of L-BFGS-B.

The Bessel functions $K_p(z)$ of the likelihood and of the GIG moments are computed on the log scale; when `besselK()` overflows (large order p, that is small ν, and small z), $\ln K_p(z)$ is computed exactly by the upward recurrence on the order $K_{p+1}(z) = K_{p-1}(z) + (2p/z)K_p(z)$.

Confidence intervals and estimation failures

With `ci = TRUE`, the function also returns asymptotic confidence intervals. Let $\hat{\psi}$ be the MLE and ℓ the log-likelihood. The observed Fisher information $J = -\partial^2\ell/\partial\psi\partial\psi^\top$ is computed at $\hat{\psi}$ by numerical differentiation (`stats::optimHess()`), and the standard errors are the square roots of the diagonal of J^{-1}. With z the quantile of order $(1 + level)/2$ of the standard normal distribution, the interval of a parameter ψ_k is

- $\hat{\psi}_k \pm z\, se_k$ for a real parameter (`mu` of the Wiener process, `theta` of the variance-gamma process, `b` and `m` of the Meixner process);
- $\hat{\psi}_k \exp(\pm z\, se_k/\hat{\psi}_k)$ for a positive parameter: the interval $\ln\hat{\psi}_k \pm z\, se_k/\hat{\psi}_k$ for $\ln\psi_k$, transformed back, which always stays positive.

These intervals are asymptotic: their coverage is close to `level` when the number of increments is large. The unbiased estimates `s2` (Wiener) and `eta_unbiased` (inverse Gaussian) are given without interval. When J is not positive definite (estimate close to a boundary of the parameter space), the bounds are `NA`.

When the maximum likelihood estimator does not exist for the data (for instance all the speeds $\Delta x_j/\Delta t_j$ equal, or a maximum at a boundary of the parameter space), the function does not stop: it gives a warning with the reason and returns `NA` estimates, with the reason in the attribute `"message"`. Invalid data (times not increasing, decreasing levels for an increasing process, too few observations) still give an error.

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See Also

`wiener_mle()`, `meixner_mle()`

Examples

```
# 1. One trajectory inspected every 2 time units up to 400
#    (theta = 0.5, sigma = 1.5, nu = 2)
t <- seq(0, 400, by = 2)
set.seed(1)
x <- DegradSim::sim_vg(t, theta = 0.5, sigma = 1.5, nu = 2)
vg_mle(x)
```

```
# The same trajectory given as two vectors: degradation levels, then times
vg_mle(x[, "X1"], t)

# 2. Twenty trajectories with the same inspection times
t <- seq(0, 40, by = 2)
x <- DegradSim::sim_vg(t, theta = 0.5, sigma = 1.5, nu = 2, n_traj = 20)
vg_mle(x)

# 3. Twenty trajectories, each with its own inspection times
plan <- DegradSim::inspection_times("dec-inc", n = 15:34, n_traj = 20, tmax = 40)
x <- DegradSim::sim_vg(plan, theta = 0.5, sigma = 1.5, nu = 2)
vg_mle(x)

# 4. Estimates with 95 % confidence intervals
vg_mle(x, ci = TRUE)

# 5. Increasing data, here from a gamma process: the likelihood is maximal
#    at the gamma limit (sigma = 0): warning suggesting gamma_mle(), NA estimates
y <- DegradSim::sim_gamma(seq(0, 100, by = 1), a = 2, b = 3)
vg_mle(y)
gamma_mle(y)
```

`wiener_mle` *Maximum likelihood estimation for the Wiener process*

Description

Computes the maximum likelihood estimator (MLE) of the drift μ and the variance σ^2 of the Wiener process $X_t = \mu t + \sigma B_t$, where B is a standard Brownian motion, from one or several trajectories observed at possibly irregular inspection times. The MLE is obtained in closed form by maximizing the likelihood directly, without numerical optimization.

Usage

```
wiener_mle(x, t = NULL, ci = FALSE, level = 0.95)
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels. The increments can also be given directly with `increments_data()`.
`t`	Inspection times of the trajectory, strictly increasing. Not used when `x` is a list.
`ci`	`FALSE` (default) for the point estimates only; `TRUE` to add asymptotic confidence intervals (see the section below).
`level`	Confidence level of the intervals, between 0 and 1 (default 0.95). Used only when `ci = TRUE`.

Value

A named vector with the maximum likelihood estimates `mu` $= \hat{\mu}$ and `sigma2` $= \hat{\sigma}^2$, and the unbiased estimate `s2` $= s^2$ of σ^2.

With `ci = TRUE`, a data frame with one row per parameter and the columns `parameter`, `estimate`, `lower` and `upper` (bounds of the confidence interval). When the estimator does not exist, the estimates (and bounds) are `NA`, with a warning.

Model and data

The increments are independent and $X_t - X_s \sim \mathcal{N}(\mu(t-s), \sigma^2(t-s))$ for $s < t$; they can be negative. The trajectories are independent and have the same law; each one is observed at its own inspection times. Let $(\Delta t_j, \Delta x_j)$, $j = 1, \ldots, m$, be the durations and degradation increments between two consecutive inspections of a same trajectory, gathered over all the trajectories, $T = \sum_j \Delta t_j$, $S = \sum_j \Delta x_j$ and $r_j = \Delta x_j / \Delta t_j$ (speeds). The likelihood is a product over all these independent increments, so the formulas below hold for one or several trajectories.

Likelihood

The increments being independent and Gaussian, the log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{m}{2}\ln(2\pi) - \frac{1}{2}\sum_j \ln \Delta t_j - \frac{m}{2}\ln\sigma^2 - \frac{Q(\mu)}{2\sigma^2}, \qquad Q(\mu) = \sum_j \frac{(\Delta x_j - \mu \Delta t_j)^2}{\Delta t_j}.$$

Maximum likelihood estimator

1. Completing the square gives $Q(\mu) = T(\mu - \hat{\mu})^2 + \hat{Q}$, with $\hat{\mu} = S/T$ and $\hat{Q} = Q(\hat{\mu}) = \sum_j \Delta t_j (r_j - \hat{\mu})^2$.
2. For every σ^2, ℓ is therefore maximal in μ at $\hat{\mu} = S/T$, the speeds averaged with weights $\Delta t_j / T$.
3. The remaining function $-\frac{m}{2}\ln\sigma^2 - \hat{Q}/(2\sigma^2)$ is maximal at $\hat{\sigma}^2 = \hat{Q}/m$.
4. The MLE exists and is unique if and only if $\hat{Q} > 0$, that is if the speeds r_j are not all equal, which has probability one as soon as $m \geq 2$. Otherwise (in particular if $m = 1$) the likelihood is unbounded (`NA` with a warning).

With one trajectory, $\hat{\mu}$ only depends on the first and last observations: the intermediate inspections only bring information on σ^2.

Bias

Writing the standardized increments as a Gaussian vector and applying an orthogonal rotation shows that, for $m \geq 2$,

$$\hat{\mu} \sim \mathcal{N}(\mu, \sigma^2/T), \qquad \hat{Q}/\sigma^2 \sim \chi^2_{m-1},$$

and that $\hat{\mu}$ and $\hat{Q}$ are independent. Hence $\hat{\mu}$ is unbiased, while $E[\hat{\sigma}^2] = \frac{m-1}{m}\sigma^2$: $\hat{\sigma}^2$ is biased downwards. The function also returns its unbiased version

$$s^2 = \frac{\hat{Q}}{m-1} = \frac{m}{m-1}\hat{\sigma}^2.$$

Confidence intervals and estimation failures

With `ci = TRUE`, the function also returns asymptotic confidence intervals. Let $\hat{\psi}$ be the MLE and ℓ the log-likelihood. The observed Fisher information $J = -\partial^2\ell/\partial\psi\partial\psi^\top$ is computed at $\hat{\psi}$ by numerical differentiation (`stats::optimHess()`), and the standard errors are the square roots of the diagonal of J^{-1}. With z the quantile of order $(1+level)/2$ of the standard normal distribution, the interval of a parameter ψ_k is

- $\hat{\psi}_k \pm z\, se_k$ for a real parameter (`mu` of the Wiener process, `theta` of the variance-gamma process, `b` and `m` of the Meixner process);
- $\hat{\psi}_k \exp(\pm z\, se_k/\hat{\psi}_k)$ for a positive parameter: the interval $\ln\hat{\psi}_k \pm z\, se_k/\hat{\psi}_k$ for $\ln\psi_k$, transformed back, which always stays positive.

These intervals are asymptotic: their coverage is close to `level` when the number of increments is large. The unbiased estimates `s2` (Wiener) and `eta_unbiased` (inverse Gaussian) are given without interval. When J is not positive definite (estimate close to a boundary of the parameter space), the bounds are `NA`.

When the maximum likelihood estimator does not exist for the data (for instance all the speeds $\Delta x_j/\Delta t_j$ equal, or a maximum at a boundary of the parameter space), the function does not stop: it gives a warning with the reason and returns `NA` estimates, with the reason in the attribute `"message"`. Invalid data (times not increasing, decreasing levels for an increasing process, too few observations) still give an error.

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See Also

`gamma_mle()`, `ig_mle()`, `vg_mle()`, `meixner_mle()`

Examples

```
# 1. One trajectory inspected every time unit up to 50 (mu = 0.5, sigma2 = 2.25)
t <- seq(0, 50, by = 1)
set.seed(1)
x <- DegradSim::sim_wiener(t, mu = 0.5, sigma = 1.5)
wiener_mle(x)

# The same trajectory given as two vectors: degradation levels, then times
wiener_mle(x[, "X1"], t)

# 2. One trajectory with irregular inspections, more and more frequent
t <- DegradSim::inspection_times("increasing", n = 50, tmax = 50)$time
x <- DegradSim::sim_wiener(t, mu = 0.5, sigma = 1.5)
wiener_mle(x)

# 3. Ten trajectories with the same inspection times
t <- seq(0, 50, by = 5)
x <- DegradSim::sim_wiener(t, mu = 0.5, sigma = 1.5, n_traj = 10)
```

```
wiener_mle(x)

# 4. Ten trajectories, each with its own inspection times
plan <- DegradSim::inspection_times("decreasing", n = 5:14, n_traj = 10, tmax = 50)
x <- DegradSim::sim_wiener(plan, mu = 0.5, sigma = 1.5)
wiener_mle(x)

# 5. Estimates with 95 % confidence intervals
wiener_mle(x, ci = TRUE)

# 6. Your own data: a list of data frames, one per trajectory,
#    with the times in the first column and the levels in the second
liste <- list(data.frame(t = c(0, 10, 20, 30), X = c(0, 4.2, 11.5, 13.9)),
              data.frame(t = c(0, 15, 30), X = c(0, 9.1, 12.8)))
wiener_mle(liste)

# sigma2 is the MLE and s2 its unbiased version: with few increments,
# here 3, they differ clearly
wiener_mle(c(0, 1.2, 0.7, 2.5), c(0, 1, 2, 3))
```

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